FUNDAMENTALS OF DATA SCIENCE EXPERIMENT

1)a) ANALYSE THE TREND OF DATA SCIENCE JOB POSTINGS OVER THE LAST DECADE

CODE :

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}


df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

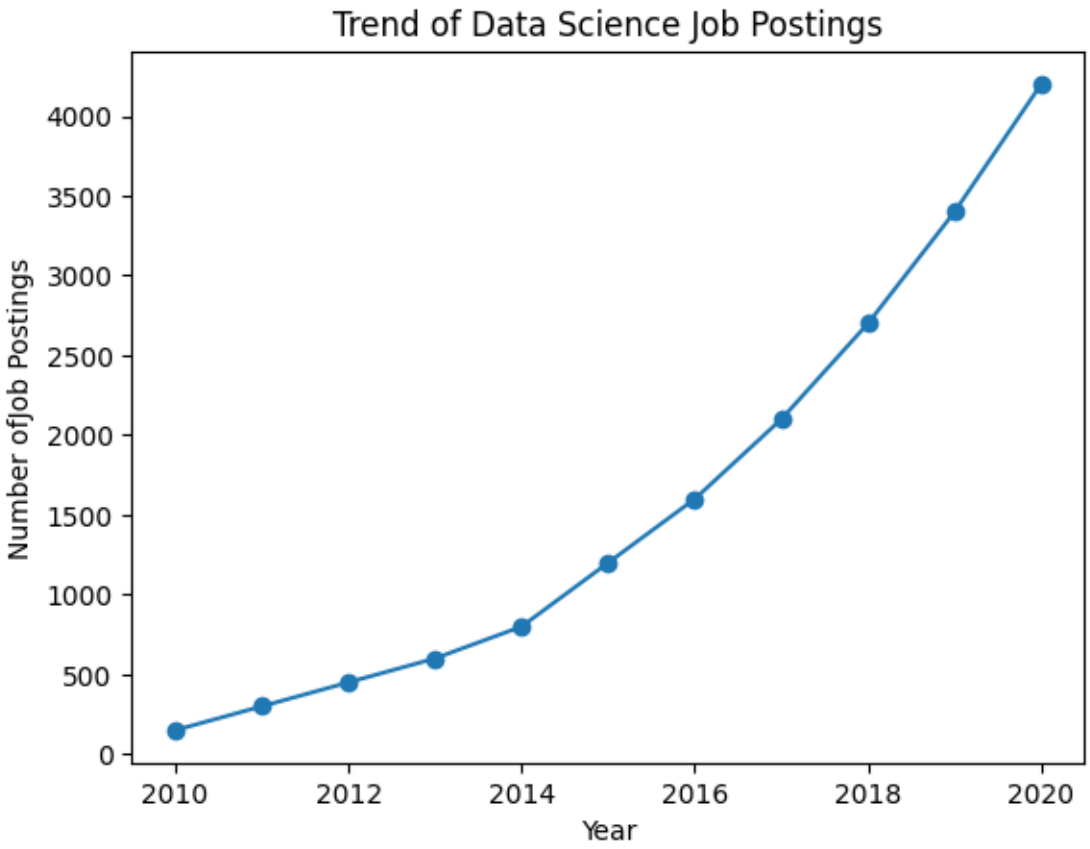
plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

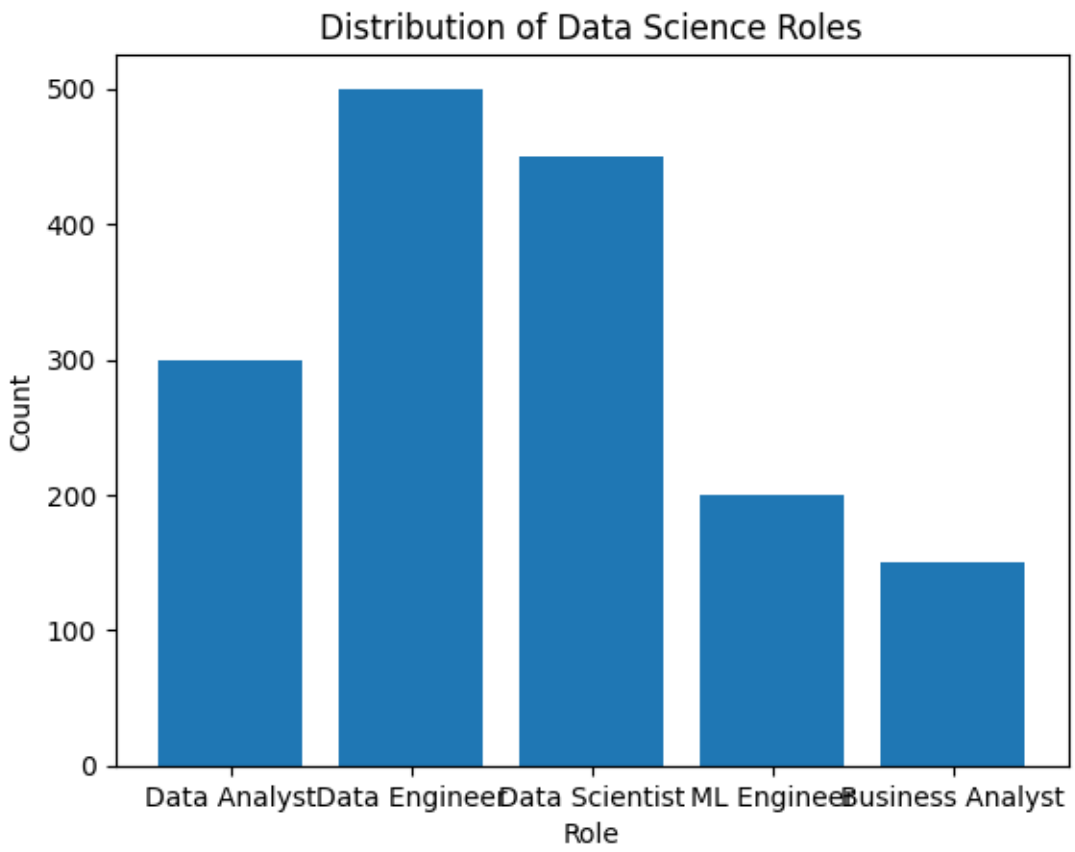
plt.ylabel('Number ofJob Postings')

plt.show()
```

OUTPUT :



1)b) ANALYSE AND VISUALIZE THE DISTRIBUTION OF VARIOUS DATA SCIENCE ROLES (DATA ANALYST, DATA ENGINEER, DATA SCIENTIST, ETC.) FROM A DATASET.



1)C) CONDUCT AN EXPERIMENT TO DIFFERENTIATE STRUCTURED , UN-STRUCTURED AND SEMI STRUCTURED DATA BASED ON DATA SETS GIVEN.

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image,or a video file.

Semi-structured Data:

{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

1)D) CONDUCT AN EXPERIMENT TO ENCRYPT AND DECRYPT GIVEN SENSITIVE DATA.

Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABqqNC3mog9w8kk7lDsdP8JmDWue0iQIreHYTP6f3gM5A0U9LQl9-kAvpC9KKQZm0zVOCsiya1AysdzVoYp0qggczryK0X8LJhM3f-Z_kFZAZNS7MTQhTlznWY9_1vs8eLKhJVE'

Decrypted Data: b'Rajalakshmi Engineering College.'